

CBSE Class 10 Mathematics — Some Applications of Trigonometry — Solutions

1. 1. A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . The height of the tower is:

Answer: (C)

Solution:

1. Let the height of the tower be h .
2. In the right triangle, $\tan(60^\circ) = \frac{\text{height}}{\text{base}}$.
3. Substitute the values: $\sqrt{3} = \frac{h}{15}$.
4. Solve for h .

Derivation:

$$\begin{aligned}\tan(60^\circ) &= \frac{h}{15} \\ \implies \sqrt{3} &= \frac{h}{15} \\ \implies h &= 15\sqrt{3} \text{ m}\end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

2. 2. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 30° . The length of the string is:

Answer: (D)

Solution:

1. Let L be the length of the string.
2. In the right triangle, $\sin(30^\circ) = \frac{\text{height}}{\text{hypotenuse}}$.
3. Substitute the values: $\frac{1}{2} = \frac{60}{L}$.
4. Solve for L .

Derivation:

$$\begin{aligned}\sin(30^\circ) &= \frac{60}{L} \\ \implies \frac{1}{2} &= \frac{60}{L} \\ \implies L &= 120 \text{ m}\end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

3. 3. The angle of elevation of the top of a pole from a point on the ground 20 m away from the foot of the pole is 45° . The height of the pole is:

Answer: (B)

Solution:

1. Let h be the height of the pole.
2. Use the relation $\tan(45^\circ) = \frac{h}{20}$.
3. Since $\tan(45^\circ) = 1$, we have $1 = \frac{h}{20}$.

Derivation:

$$\begin{aligned}\tan(45^\circ) &= \frac{h}{20} \\ \implies 1 &= \frac{h}{20} \\ \implies h &= 20 \text{ m}\end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

4. 4. If the length of the shadow of a tower is $\sqrt{3}$ times the height of the tower, then the angle of elevation of the sun is:

Answer: (A)

Solution:

1. Let the height of the tower be h . Then the shadow length is $h\sqrt{3}$.
2. Let θ be the angle of elevation.
3. Use $\tan(\theta) = \frac{h}{h\sqrt{3}}$.
4. Simplify to find θ .

Derivation:

$$\begin{aligned}\tan(\theta) &= \frac{h}{h\sqrt{3}} = \frac{1}{\sqrt{3}} \\ \implies \theta &= 30^\circ\end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

5. 5. An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45° . The height of the chimney is:

Answer: (C)

Solution:

1. Height of observer = 1.5 m. Distance = 28.5 m.
2. Height of chimney above observer's eye level = $28.5 \times \tan(45^\circ) = 28.5$ m.
3. Total height = $28.5 + 1.5$.

Derivation:

$$H = 28.5 + 1.5 = 30 \text{ m}$$

Some Applications of Trigonometry — Heights and Distances

6. 6. Assertion (A): The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower of height $30\sqrt{3}$ m, is 60° . Reason (R): In a right-angled triangle, $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.

Answer: (A)

Solution:

1. Identify the height (opposite) = $30\sqrt{3}$ and base (adjacent) = 30.
2. Use the formula $\tan(\theta) = \frac{30\sqrt{3}}{30} = \sqrt{3}$.
3. Since $\tan(60^\circ) = \sqrt{3}$, the angle is 60° .

Derivation:

$$\begin{aligned}\tan(\theta) &= \frac{30\sqrt{3}}{30} = \sqrt{3} \\ \implies \theta &= 60^\circ\end{aligned}$$

Some Applications of Trigonometry — Angle of Elevation

7. **7.** Assertion (A): If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the Sun is 45° . Reason (R): The angle of elevation is defined only when the object is taller than the observer.

Answer: (C)

Solution:

1. Let height = h and shadow = h .
2. $\tan(\theta) = \frac{h}{h} = 1$.
3. $\theta = 45^\circ$. Assertion is true.
4. Reason is false because angle of elevation is defined by the line of sight relative to the horizontal, regardless of relative height.

Derivation:

$$\begin{aligned}\tan(\theta) &= \frac{\text{height}}{\text{shadow}} = \frac{h}{h} = 1 \\ \implies \theta &= 45^\circ\end{aligned}$$

Some Applications of Trigonometry — Angle of Elevation

8. **8.** Assertion (A): The angle of depression of an object from a point h meters above the ground is always equal to the angle of elevation of that point from the object on the ground. Reason (R): The angle of elevation and depression are alternate interior angles for parallel horizontal lines.

Answer: (A)

Solution:

1. By the properties of parallel lines, the line of sight creates alternate interior angles.
2. *Thus, angle of elevation = angle of depression.*

Derivation:

$$\angle \text{elevation} = \angle \text{depression} \text{ (Alternate Interior Angles)}$$

Some Applications of Trigonometry — Angle of Depression

9. **9.** Assertion (A): If a ladder of length 10 m reaches a wall at a height of 5 m, the angle made by the ladder with the ground is 30° . Reason (R): $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$.

Answer: (A)

Solution:

1. Opposite = 5, Hypotenuse = 10.
2. $\sin(\theta) = \frac{5}{10} = 0.5$.
3. $\theta = 30^\circ$.

Derivation:

$$\begin{aligned}\sin(\theta) &= \frac{5}{10} = \frac{1}{2} \\ \implies \theta &= 30^\circ\end{aligned}$$

Some Applications of Trigonometry — Trigonometric Ratios

10. **10.** Assertion (A): As an observer moves closer to a tower, the angle of elevation of its top increases. Reason (R): $\tan(\theta)$ increases as θ increases from 0° to 90° .

Answer: (A)

Solution:

1. In $\tan(\theta) = \frac{h}{x}$, as x (distance) decreases, $\tan(\theta)$ increases.
2. As $\tan(\theta)$ increases, θ must increase.

Derivation:

$$\text{As } x \rightarrow 0, \tan(\theta) = \frac{h}{x} \rightarrow \infty, \text{ so } \theta \rightarrow 90^\circ$$

Some Applications of Trigonometry — Angle of Elevation

11. **11.** A ladder 15 m long makes an angle of 60° with the wall. Find the height of the point where the ladder touches the wall.

Answer:

$$7.5m$$

Solution:

1. Identify the triangle formed by the ladder (hypotenuse), the wall (adjacent side to 60 degrees), and the ground.
2. Use the trigonometric ratio $\cos(60^\circ) = \frac{\text{adjacent}}{\text{hypotenuse}}$ to find the height.

Derivation:

$$\begin{aligned}\cos(60^\circ) &= \frac{h}{15} \\ \implies \frac{1}{2} &= \frac{h}{15} \\ \implies h &= 7.5 \text{ m}\end{aligned}$$

Some Applications of Trigonometry — Trigonometric Ratios

- 12. 12.** The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30° . Find the height of the tower.

Answer:

$$10\sqrt{3}m$$

Solution:

1. Use the tangent ratio $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$ where $\theta = 30^\circ$.
2. Substitute the given values and solve for the height h .

Derivation:

$$\begin{aligned}\tan(30^\circ) &= \frac{h}{30} \\ \implies \frac{1}{\sqrt{3}} &= \frac{h}{30} \\ \implies h &= \frac{30}{\sqrt{3}} = 10\sqrt{3} \text{ m}\end{aligned}$$

Some Applications of Trigonometry — Angle of Elevation

- 13. 13.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string.

Answer:

$$40\sqrt{3}m$$

Solution:

1. Represent the situation as a right triangle where the height is the opposite side and string length is the hypotenuse.
2. Use $\sin(60^\circ) = \frac{\text{height}}{\text{length}}$ to solve for the length.

Derivation:

$$\begin{aligned}\sin(60^\circ) &= \frac{60}{L} \\ \implies \frac{\sqrt{3}}{2} &= \frac{60}{L} \\ \implies L &= \frac{120}{\sqrt{3}} = 40\sqrt{3} \text{ m}\end{aligned}$$

Some Applications of Trigonometry — Real-world problems

- 14. 14.** A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Answer:

$$19\sqrt{3}m$$

Solution:

1. Determine the effective height of the building relative to the boy's eye level:
 $30 - 1.5 = 28.5 \text{ m}.$

2. Use $\cot(30^\circ)$ and $\cot(60^\circ)$ to find the difference in ground distances.

Derivation:

$$d = 28.5(\cot(30^\circ) - \cot(60^\circ)) = 28.5(\sqrt{3} - \frac{1}{\sqrt{3}}) = 28.5(\frac{2}{\sqrt{3}}) = \frac{57}{\sqrt{3}} = 19\sqrt{3} \text{ m}$$

Some Applications of Trigonometry — Angle of Elevation

- 15. 15.** From a point on the ground, the angles of elevation of the bottom and top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.

Answer:

$$20(\sqrt{3} - 1)m$$

Solution:

1. Let h be the height of the tower. The building height is 20.
2. Use $\tan(45^\circ)$ to find the distance from the point to the building, then use $\tan(60^\circ)$ for the total height.

Derivation:

$$\begin{aligned} \text{Distance } x &= \frac{20}{\tan(45^\circ)} = 20. \text{ Total height } (20 + h) = x \tan(60^\circ) = 20\sqrt{3}. \\ \implies h &= 20\sqrt{3} - 20 = 20(\sqrt{3} - 1) \text{ m} \end{aligned}$$

Some Applications of Trigonometry — Multiple Angles of Elevation

- 16. 16.** A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . Find the height of the tower.

Answer:

$$15\sqrt{3} \text{ m}$$

Solution:

1. Let h be the height of the tower and $d = 15$ m be the distance from the foot.
2. Use the tangent ratio: $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.
3. Substitute the values: $\tan(60^\circ) = \frac{h}{15}$.
4. Solve for h : $h = 15 \times \sqrt{3}$.

Derivation:

$$\begin{aligned} \tan(60^\circ) &= \frac{h}{15} \\ \implies \sqrt{3} &= \frac{h}{15} \\ \implies h &= 15\sqrt{3} \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

17. **17.** A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Answer:

$$19\sqrt{3} \text{ m}$$

Solution:

1. Height of building above boy's eyes = $30 - 1.5 = 28.5$ m.
2. Let x_1 be the distance at 30° and x_2 be the distance at 60° .
3. Calculate $x_1 = 28.5 \cot(30^\circ)$ and $x_2 = 28.5 \cot(60^\circ)$.
4. Distance walked = $x_1 - x_2 = 28.5(\sqrt{3} - \frac{1}{\sqrt{3}}) = 28.5(\frac{2}{\sqrt{3}}) = 19\sqrt{3}$.

Derivation:

$$\begin{aligned} x_1 &= 28.5\sqrt{3}, x_2 = \frac{28.5}{\sqrt{3}} \\ \Rightarrow \text{dist} &= 28.5\left(\frac{3-1}{\sqrt{3}}\right) = \frac{57}{\sqrt{3}} = 19\sqrt{3} \end{aligned}$$

Some Applications of Trigonometry — Two angles of elevation

18. **18.** From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.

Answer:

$$7(1 + \sqrt{3}) \text{ m}$$

Solution:

1. Let the building height $AB = 7$ m. The tower height $CD = h$.
2. From the triangle at the base, the distance between building and tower is $7 \cot(45^\circ) = 7$ m.
3. In the upper triangle, the height above the building level is $h' = 7 \tan(60^\circ) = 7\sqrt{3}$.
4. Total height $h = 7 + 7\sqrt{3}$.

Derivation:

$$h = 7 + 7 \tan(60^\circ) = 7(1 + \sqrt{3})$$

Some Applications of Trigonometry — Angle of elevation and depression

19. **19.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming that there is no slack in the string.

Answer:

$$40\sqrt{3} \text{ m}$$

Solution:

1. Height $h = 60$ m, angle $\theta = 60^\circ$. Let string length be L .

2. Use sine ratio: $\sin(60^\circ) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{60}{L}$.
3. Substitute $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.
4. Solve: $L = \frac{60 \times 2}{\sqrt{3}} = \frac{120}{\sqrt{3}} = 40\sqrt{3}$.

Derivation:

$$\begin{aligned}\sin(60^\circ) &= \frac{60}{L} \\ \implies \frac{\sqrt{3}}{2} &= \frac{60}{L} \\ \implies L &= \frac{120}{\sqrt{3}} = 40\sqrt{3}\end{aligned}$$

Some Applications of Trigonometry — Sine ratio application

- 20. 20.** The angle of elevation of the top of a chimney from the top of a tower is 30° and the angle of depression of the foot of the chimney from the top of the tower is 60° . If the tower is 40 m high, find the height of the chimney.

Answer:

$$160/3 \text{ m}$$

Solution:

1. Let h be the height of the chimney above the tower level and 40 m be the tower height.
2. Distance $d = 40 \cot(60^\circ) = \frac{40}{\sqrt{3}}$.
3. Height $h = d \tan(30^\circ) = \frac{40}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{40}{3}$.
4. Total height $= 40 + \frac{40}{3} = \frac{160}{3}$ m.

Derivation:

$$H = 40 + \frac{40}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}} = 40 + \frac{40}{3} = \frac{160}{3}$$

Some Applications of Trigonometry — Application with two points

- 21. 21.** A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.

Answer:

$$3 \text{ seconds}$$

Solution:

1. Let h be the height of the tower and v be the uniform speed of the car.
2. Let A be the top of the tower and BC be the distance covered by the car in 6 seconds.
3. In $\triangle ABC$ (with angle 60° at the car), $\tan 60^\circ = h/CD$.
4. In $\triangle ABD$ (with angle 30° at the car), $\tan 30^\circ = h/BD$.

5. Relate the distances using the speed and time formula $d = v \times t$.
6. Solve for the time t taken to travel the remaining distance CD .

Derivation:

$$\Rightarrow x + d = h\sqrt{3}. \text{Substituting } x : d = h\sqrt{3} - h/\sqrt{3} = 2h/\sqrt{3}. \text{Since distance discovered in 6s, speed}$$

Some Applications of Trigonometry — Angle of depression

- 22. 22.** The angle of elevation of the top of a tower from a point on the ground is 30° . On walking 40 metres towards the tower, the angle of elevation changes to 60° . Find the height of the tower and the distance of the initial point from the tower.

Answer:

$$\text{Height} = 20\sqrt{3}m, \text{Distance} = 20\sqrt{3} + 40m$$

Solution:

1. Let the height of the tower be h and the distance from the second point be x .
2. Set up the equation for the first point: $\tan 30^\circ = h/(x + 40)$.
3. Set up the equation for the second point: $\tan 60^\circ = h/x$.
4. Express h in terms of x from the second equation.
5. Substitute into the first equation to solve for x and then h .

Derivation:

$$\begin{aligned} 1) h/x &= \sqrt{3} \\ \Rightarrow h &= x\sqrt{3} \\ 2) h/(x + 40) &= 1/\sqrt{3} \\ \Rightarrow x\sqrt{3}/(x + 40) &= 1/\sqrt{3} \\ \Rightarrow 3x &= x + 40 \\ \Rightarrow 2x &= 40 \\ \Rightarrow x &= 20. \text{Height } h = 20\sqrt{3}m. \text{Initial distance} = 20 + 40 = 60m. \end{aligned}$$

Some Applications of Trigonometry — Angle of elevation

- 23. 23.** A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 metres. At a point on the plane, the angle of elevation of the bottom and top of the flagstaff are 30° and 60° respectively. Find the height of the tower.

Answer:

$$3\text{metres}$$

Solution:

1. Let h be the height of the tower and d be the distance from the point to the base.
2. Write the equation for the bottom of the flagstaff: $\tan 30^\circ = h/d$.

3. Write the equation for the top of the flagstaff: $\tan 60^\circ = (h + 6)/d$.
4. Solve the system of equations for h .

Derivation:

$$\begin{aligned}
 d = h / \tan 30^\circ = h\sqrt{3}. \text{Substituting into second equation : } \sqrt{3} &= (h + 6)/(h\sqrt{3}) \\
 \implies 3h &= h + 6 \\
 \implies 2h &= 6 \\
 \implies h &= 3\text{m}.
 \end{aligned}$$

Some Applications of Trigonometry — Height of tower with flagstaff

- 24. 24.** From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.

Answer:

$$7(1 + \sqrt{3})\text{metres}$$

Solution:

1. Let the building be $AB = 7$ m and the tower be CD .
2. Draw a horizontal line from A to meet CD at E . Thus $AE = BD$.
3. In $\triangle ABD$, since the angle of depression is 45° , $BD = AB = 7$ m.
4. In $\triangle ACE$, $\tan 60^\circ = CE/AE$.
5. Calculate CE and add $ED = AB = 7$ m to get the total height.

Derivation:

$$AE = BD = 7. \text{In } \triangle ACE, CE = AE \tan 60^\circ = 7\sqrt{3}. \text{Total height } CD = CE + ED = 7\sqrt{3} + 7 = 7(\sqrt{3} + 1)$$

Some Applications of Trigonometry — Combined elevation and depression